Mandala Vishista Reddy

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.52
Sri Chaitanya	Class XII (Telangana State Board)	2023	98.30%
Delhi Public School, Warangal	Class X (CBSE)	2021	97.60%

SKILLS

Languages: Python, C/C++, SQL

ML/DL: PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, YOLO, U-Net, Diffusion Models

LLM/GenAI: LangChain, LangGraph, RAG, FAISS, ChromaDB, Hugging Face, Pydantic

Deployment: ONNX, TensorRT, INT8 Quantization, FastAPI, Docker, AWS (EC2, ECR), GitHub Actions

Tools: Git, MySQL, Jupyter, VS Code, Excel

Libraries: Numpy, Pandas, Matplotlib, Seaborn, Scikit learn, Pytorch, OpenCV, Pydantic, Langchain, Langgraph.

Core CS Fundamentals: DBMS, OS

WORK EXPERIENCE

AI Engineering Intern | DeepGrid Semi Pvt Ltd

- Fine-tuned **YOLO26n** on FLIR ADAS v2 (15 classes) via two-stage IR to IR+RGB transfer learning, achieving **mAP50 38.2% / mAP50-95 22.3%**.
- Applied 20% structured pruning and INT8 QAT, cutting model size by **48% (5.39MB to 2.81MB)** for edge deployment.
- Built an ONNX to TensorRT INT8 pipeline (onnx, calibration preprocessing), achieving **2.1ms/image** GPU latency.
- Designed **EgoLaneNet**, a 3.2M-param MobileNetV2 lane detector: **72.1% val accuracy** on CULane, then fine-tuned on TuSimple, BDD100K and CurveLanes.
- Deployed **CLRerNet on Rockchip RK3588 NPU** via ONNX graph surgery and INT8 quantization: **60MB to 15MB at ~80 FPS**

PROJECTS

AhvisGPT | Python, LangGraph, Gemini, FastAPI, ChromaDB, Docker, AWS

- Built an **agentic AI assistant** using LangGraph and Gemini, with a **5-tool workflow** that picks tools by intent and uses **RAG over 6 document formats**.
- Developed a **FastAPI backend** with SSE streaming and persistent memory across 3 LLMs, deployed through **CI/CD (GitHub Actions to AWS ECR/EC2).**

AI-Powered News Research Tool | LangChain, OpenAI, FAISS, Streamlit

- Built a **RAG pipeline** that takes news article URLs, chunks and embeds their content into a **FAISS vector store**, and retrieves the most relevant passages.
- Created a **Streamlit app** that answers plain-English questions across all articles in one reply, **citing the exact source text** behind each answer.

Orographic Precipitation Diffusion | TensorFlow, 3D U-Net, DDPM, Satellite Data (Ongoing)

- Built a **multi-modal 3D U-Net** fusing IMERG satellite rainfall, DEM terrain and atmospheric data to **bias-correct**

rainfall over the Western Ghats, evaluated with RMSE and KL/JS divergence.
- Designing a **terrain-conditioned diffusion model (DDPM)** with a physics-informed loss, extending PrecipDiff (AAAI 2025) to fix **windward/leeward rainfall errors**.

RELEVANT COURSES

Computer Science: Introduction to Programming.

Math: Probability, Statistics, Linear Algebra, Calculus, Differential Equations, Complex Variables, Transform Techniques

Electrical Engineering: Basic Electrical Engineering

Economics: Macroeconomics, Public Finance and Public Policy.

POSITIONS OF RESPONSIBILITY

Cohort Lead, OCS: Leading a team of **15 coordinators** to pitch to companies and track placement progress for Civil department. Reached out to **30+ companies** and coordinated internship and Placement drives between recruiters and **150+ students**.

Events Coordinator:

- **EBSB**: Organised cultural events throughout the year promoting the heritage of our country.
- **NSS**: Organised **5+ community service drives** (e.g., blood donation, clean India Drive) with **50+ volunteers**.

Core Member, Shuffle Dance Club: Choreographed and performed at many institute events and Inter IIT; helped train new members.

EXTRACURRICULAR

Sports & Cultural:

- 1st Prize, Freshers' **Badminton Competition**, IIT Hyderabad (2023)
- 2nd Prize, **Cricket**, Diesta departmental fest (2025)
- **Karate Black Belt**; 8 medals, incl. Gold in Kata at JKS National Invitational Championship (2015)
- **Inter IIT Cultural Meet**, Group **Dance team**: IIT Kharagpur (2023), IIT Patna (2024)
- 2nd Prize, **Street Battle**, Milan fest (2023)